

BLUESTOCK

Mutual Fund Analytics Capstone

FINAL PROJECT REPORT

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Tools: Python · SQL · SQLite · Power BI · Pandas · Plotly · NumPy

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1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone is a comprehensive, end-to-end data analytics project developed to transform raw mutual fund data into actionable investment intelligence. Executed over seven structured phases, the project encompasses the full data analytics lifecycle — from raw data ingestion and pipeline engineering through exploratory analysis, quantitative performance modelling, and interactive business intelligence dashboarding.

The project analysed 40 mutual fund schemes across 10 fund houses, processing over 46,000 NAV observations and 32,000 investor transactions. Advanced financial metrics — including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk, and Conditional Value at Risk — were computed to provide a rigorous quantitative evaluation of each scheme. A composite fund scorecard was developed to rank all 40 schemes on a unified performance framework.

Key industry findings revealed that SBI Mutual Fund led the industry in Assets Under Management, SIP participation demonstrated a sustained upward trend from 2022 through 2025, and small-cap funds carried the highest downside risk as measured by VaR and CVaR. A four-page interactive Power BI dashboard was developed to support real-time monitoring and stakeholder reporting.

40 Mutual Fund Schemes	10 Fund Houses Analysed	46,000+ NAV Observations	32,778 Investor Transactions
5 Risk Categories	10 Datasets Used	4 Dashboard Pages	10 SQL Queries Built

Project Verdict: All eight project objectives were successfully met. All seven deliverables were completed and are production-ready. The pipeline executes end-to-end without errors, and the dashboard loads correctly from the published .pbix file.

2. Project Objectives

The capstone was designed around six primary analytical objectives, each targeting a distinct dimension of mutual fund intelligence:

Objective 1	Multi-Category Performance Analysis Evaluate and compare mutual fund performance across equity, debt, hybrid, and thematic fund categories using quantitative return and risk metrics.
Objective 2	Historical NAV Trend Evaluation Track and visualize daily NAV movements for all 40 schemes across the 2022–2026 period, highlighting the 2023 bull run and the 2024 market correction.
Objective 3	Investor Behaviour & SIP Analysis Study transaction patterns, SIP inflow trends, investor demographics, and geographic distribution of retail mutual fund participation.
Objective 4	Benchmark Comparison Assess fund performance relative to NIFTY50 and NIFTY100 benchmarks using tracking error and normalized growth analysis.
Objective 5	Quantitative Risk Modelling Compute industry-standard risk metrics — CAGR, Volatility, Sharpe Ratio, Sortino Ratio, Beta, VaR (95%), and CVaR (95%) — across all 40 schemes.
Objective 6	Business Intelligence Dashboarding Develop an interactive Power BI dashboard to consolidate performance, investor, SIP, and risk data into a single decision-support platform.

3. Data Sources & Dataset Description

The project ingests ten structured CSV datasets representing the complete mutual fund ecosystem — covering fund metadata, daily pricing, asset management, investor activity, portfolio allocation, and market benchmarks. In addition, live NAV data was retrieved programmatically via the public mfapi.in REST API.

#	Dataset	Description
01	Fund Master	Master scheme info: categories, fund houses, managers, investment limits
02	NAV History	Daily NAV records for 40 schemes — primary source for return analytics
03	AUM by Fund House	Assets Under Management aggregated by fund house and year
04	Monthly SIP Inflows	Monthly SIP contribution data tracking retail participation trends
05	Category Inflows	Net inflows and outflows segmented by fund category
06	Industry Folio Count	Total industry folio counts to measure investor base growth
07	Scheme Performance	Pre-computed returns, Sharpe ratio, alpha, beta, expense ratio
08	Investor Transactions	32,778 individual transactions: SIP, lumpsum, and redemptions
09	Portfolio Holdings	Stock-level portfolio weights and sector exposure per scheme
10	Benchmark Indices	NIFTY50 and NIFTY100 daily index data for comparative analysis

Live Data Integration: Real-time NAV data for selected fund schemes was fetched using the mfapi.in public API and stored locally, enabling up-to-date pricing alongside the historical dataset. AMFI code validation achieved a **100% match rate** across all 40 schemes.

4. ETL Pipeline Design

A production-grade ETL (Extract, Transform, Load) pipeline was engineered to automate the full data workflow from raw CSV ingestion through analytics-ready database loading. The pipeline is orchestrated by a single master script, `run_pipeline.py`, ensuring reproducible, one-command execution.

4.1 Extract Phase

Raw datasets were ingested from the `data/raw/` directory using Pandas `read_csv()` with explicit dtype mapping and encoding specification. Live NAV data was fetched via HTTP GET requests to the `mfapi.in` API using the Requests library, with AMFI codes validated against the fund master before storage.

4.2 Transform Phase — Data Cleaning

Fund Master

- Removed duplicate AMFI codes
- Validated expense ratios within permissible range (0.1%–2.5%)
- Standardised minimum SIP and lumpsum investment fields

NAV History

- Converted date column to Python datetime format
- Sorted records by AMFI code and chronological date
- Removed duplicate NAV entries
- Forward-filled missing NAV values to preserve time-series continuity
- Validated all NAV values are strictly greater than zero

Investor Transactions

- Standardised transaction types to: SIP, Lumpsum, Redemption
- Converted transaction dates to uniform datetime format
- Validated all transaction amounts are positive
- Verified KYC status field integrity

Scheme Performance

- Converted return metric columns to numeric dtype
- Validated performance indicator ranges
- Flagged anomalous records to a separate review file

Portfolio Holdings

- Validated portfolio weight totals per scheme
- Verified stock prices and market values are non-negative

4.3 Validation Framework

A dedicated validation layer ran six categories of checks across all datasets before loading: duplicate detection, missing value analysis, data type validation, business rule enforcement, range validation, and category standardisation. All datasets passed validation with zero remaining duplicates and fully standardised data types.

4.4 Load Phase — SQLite Star Schema

Cleaned datasets were loaded into a SQLite database (bluestock_mf.db) modelled as a star schema optimised for analytical query performance. SQLAlchemy and Pandas to_sql() were used for database population.

Table	Type	Rows	Description
dim_fund	Dimension	40	Scheme metadata, fund house, manager, risk category
dim_date	Dimension	—	Calendar table: year, quarter, month, day
fact_nav	Fact	46,000	Daily NAV history for all schemes
fact_transactions	Fact	32,778	Investor transaction records
fact_performance	Fact	40	Scheme-level performance metrics
fact_aum	Fact	90	Fund house AUM data by period

4.5 SQL Analytics Layer

Ten analytical SQL queries were implemented against the SQLite database to validate data integrity and generate business insights, including: Top 5 Fund Houses by AUM, Highest 5-Year Return Funds, Lowest Expense Ratio Funds, Transaction Volume by Type, Top States by Investment Amount, and KYC Status Distribution. All queries executed successfully.

5. Exploratory Data Analysis — Key Findings

A comprehensive, nine-module EDA was executed across all ten datasets using Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly). The analysis surfaces patterns in NAV movements, investor behaviour, AUM distribution, sector allocation, and geographic participation.

1	NAV Trend Analysis Daily NAV movements were visualised for all 40 schemes across the full observation window. A strong bull market rally was identified during 2023, with a notable correction phase evident across multiple schemes during 2024. Interactive Plotly charts enabled scheme-level drill-down.
2	AUM Growth Analysis AUM trends were tracked across all 10 fund houses. SBI Mutual Fund emerged as the clear industry leader by total assets. Year-on-year AUM comparison revealed accelerating growth in equity-oriented fund houses.
3	SIP Inflow Analysis Monthly SIP inflow data from 2022 to 2025 demonstrated a consistent upward trajectory in retail participation. The highest SIP inflow period was identified during the second half of 2024, reflecting growing investor confidence.
4	Category-Wise Inflow Heatmaps Inflow heatmaps across fund categories revealed that equity and flexi-cap categories consistently attracted the highest net inflows. Debt fund categories showed relative stability with lower volatility in net flows.
5	Investor Demographics Age-group distribution analysis revealed that younger investors (25–35 age bracket) formed the largest segment of the investor base. Gender participation data indicated growing female investor engagement across SIP-based investment modes.
6	Geographic Distribution State-wise transaction analysis highlighted that Tier-1 (T30) cities dominated total investment volume. However, B30 city participation showed meaningful and accelerating growth, indicating expanding retail investor reach across smaller markets.
7	Folio Growth Tracking Industry folio count data revealed significant expansion in the total investor base across the study period, with major growth milestones achieved in 2023 and 2024, consistent with rising SIP adoption.
8	Return Correlation Matrix Cross-fund return correlation analysis identified opportunities for portfolio diversification. Several small-cap funds demonstrated lower correlation with large-cap schemes, suggesting potential diversification benefits.
9	Sector Allocation Analysis Portfolio holdings were aggregated by sector to reveal concentration patterns. Financial services and technology sectors represented the dominant allocations across equity mutual fund portfolios, as illustrated by sector donut charts.

Summary of Key EDA Findings

- ✓ NAV data confirmed the 2023 bull market across equity fund categories, with corrections materialising in 2024.
- ✓ SBI Mutual Fund held the highest AUM position throughout the study period.
- ✓ SIP inflows demonstrated a sustained multi-year growth trend, reflecting expanding retail participation.
- ✓ Folio counts grew significantly in 2023–2024, indicating accelerating investor acquisition.
- ✓ Younger investor cohorts (25–35) dominated the investor demographic profile.
- ✓ B30 cities are growing in contribution, indicating geographic diversification of the investor base.
- ✓ Financial services and technology sectors represented the largest portfolio weights across equity schemes.
- ✓ Cross-fund return correlations revealed meaningful diversification potential within the analysed universe.

6. Fund Performance Analytics

A rigorous quantitative performance evaluation framework was applied across all 40 mutual fund schemes, processing approximately 46,000 NAV observations. The framework combines return-based, risk-adjusted, benchmark-relative, and predictive analytics to generate a comprehensive view of each scheme.

6.1 Daily Return Analysis

Daily returns were computed for every scheme using: $\text{daily_return} = \text{NAV}(t) / \text{NAV}(t-1) - 1$. Return distribution statistics were generated to validate data quality and characterise volatility patterns across the fund universe.

6.2 CAGR Analysis

Compounded Annual Growth Rate (CAGR) was calculated for all available periods. Schemes were ranked comparatively, identifying top performers based on long-term annualised growth. Note: Historical 5-Year CAGR computation was constrained by data availability (2022–2026); ARIMA-based forecasted 5-Year CAGR values were generated as a supplement.

6.3 Risk-Adjusted Metrics

Sharpe Ratio (risk-free rate: 6.5%) was used to rank funds by excess return per unit of total risk. The Sortino Ratio was separately computed to isolate downside risk, providing a more conservative view of risk-adjusted performance for schemes with asymmetric return distributions.

6.4 Alpha & Beta (Benchmark Regression)

Linear regression of fund daily returns against the NIFTY100 benchmark yielded Alpha (active return over benchmark), Beta (market sensitivity), and R-squared (benchmark explanation power) for every scheme. Mid-cap and emerging equity funds demonstrated the highest alpha generation.

6.5 Maximum Drawdown Analysis

Peak-to-trough decline was measured for every scheme across the full observation window. Drawdown rankings identified the highest-risk and lowest-risk schemes, providing a critical risk boundary metric for investor suitability assessment.

6.6 Composite Fund Scorecard

A weighted composite scoring model (0–100 scale) was developed to rank all 40 schemes on a single unified framework:

Metric	Weight	Rationale
3-Year CAGR Rank	30%	Primary return metric over medium-term horizon
Sharpe Ratio Rank	25%	Risk-adjusted excess return quality
Alpha Rank	20%	Active outperformance vs benchmark
Expense Ratio Rank (Inv)	15%	Lower cost = higher net investor return
Max Drawdown Rank (Inv)	10%	Capital protection and downside risk control

6.7 Benchmark Comparison & Tracking Error

Top-ranked funds were assessed against NIFTY50 and NIFTY100 through normalised growth comparisons and tracking error calculations. The tracking error metric quantified how closely each fund replicated benchmark behaviour — important for distinguishing active alpha generators from closet indexers.

6.8 Predictive Analytics — ARIMA Forecasting

ARIMA time-series models were fitted on NAV histories of top-performing funds to generate 1-year NAV forecasts, projected returns, and estimated 5-year CAGR. While forecasts are inherently directional indicators rather than guarantees, they demonstrate the project's capability to extend beyond descriptive into predictive financial analytics.

7. Advanced Analytics & Risk Metrics

Day 6 extended the project into advanced financial risk modelling and investor intelligence, implementing six analytical modules that go beyond descriptive statistics to deliver predictive risk signals, investor segmentation, and a rule-based recommendation engine.

7.1 Value at Risk (VaR) & CVaR Analysis

Historical VaR at the 95% confidence level was computed for all 40 schemes using the 5th percentile of daily return distributions. Conditional VaR (CVaR) — the average return in the worst 5% of outcomes — was computed as a more conservative tail-risk measure. Small-cap funds (SBI Small Cap, Axis Small Cap, ABSL Small Cap) exhibited the most negative VaR and CVaR values, confirming their high-risk profile.

7.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed for the top 5 funds by AUM using the formula: $(\text{Rolling Mean Return} / \text{Rolling Std Dev}) \times \sqrt{252}$. Time-series visualisations revealed periods of elevated and suppressed risk-adjusted performance, providing a dynamic view of fund quality across market cycles.

7.3 Investor Cohort Analysis

Investors were segmented by year of first transaction and analysed by average investment amount, total invested volume, and fund preference. The 2024 cohort represented the largest investment volume, with Mirae Asset Emerging Bluechip Fund emerging as the most preferred scheme among new investors.

7.4 SIP Continuity Analysis

SIP transaction gaps were evaluated for investors with six or more SIP transactions. Investors with an average gap exceeding 35 days were flagged as At-Risk. Analysis revealed that only 2.20% of eligible investors maintained healthy SIP continuity, highlighting a significant behavioural risk for fund houses relying on SIP inflows.

7.5 Sector Concentration (HHI)

The Herfindahl-Hirschman Index ($\text{HHI} = \text{sum of squared portfolio weights}$) was applied to measure portfolio concentration. Axis Bluechip Fund recorded the highest HHI among analysed funds, indicating a highly concentrated portfolio relative to peers.

7.6 Mutual Fund Recommendation Engine

A rule-based recommendation system was built to suggest funds by investor risk profile (Low / Moderate / High Risk). Ranking criteria combined Sharpe Ratio, historical performance, and risk grade. Sample high-risk recommendations included Kotak Emerging Equity Fund, ICICI Prudential Midcap Fund, and DSP Midcap Fund.

Risk Highlight: Small-cap mutual funds demonstrated the highest downside risk. Only 2.20% of SIP investors maintained healthy contribution continuity. The 2024 investor cohort drove the majority of total investment volume in the analysed dataset.

8. Power BI Dashboard Overview

A four-page interactive Power BI dashboard was developed as the primary business intelligence deliverable, consolidating all analytical outputs into a single stakeholder-ready reporting platform. The dashboard supports dynamic filtering, cross-visual interaction, and KPI monitoring.

Page 1 — Industry Overview

Displays KPI cards for Total AUM, Total SIP Inflows, Total Schemes, and Total Fund Houses. Features an AUM trend line (2022–2025) highlighting the 2023 peak and 2024 correction, plus a horizontal bar chart ranking fund houses by AUM — with SBI Mutual Fund leading at the top. An interactive year slicer enables dynamic period filtering across all visuals.

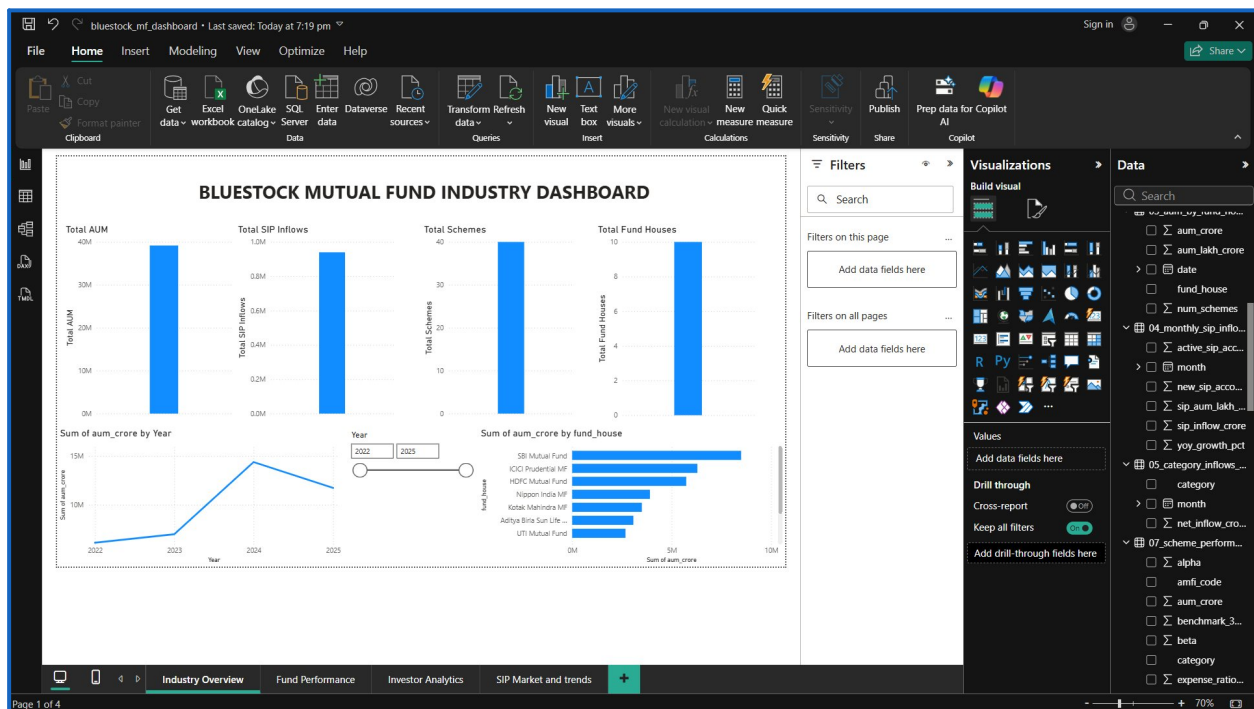


Figure: Page 1 — Industry Overview — Power BI Dashboard

Page 2 — Fund Performance

A Risk vs Return scatter plot maps all schemes by 3-Year Return against return volatility, with category-level colour coding. The Top Performing Funds scorecard table lists schemes by return and risk metrics. A multi-line NAV Trend chart shows NAV trajectories from 2022–2026 for selected schemes. Fund house, category, and plan filter slicers enable drill-down analysis.

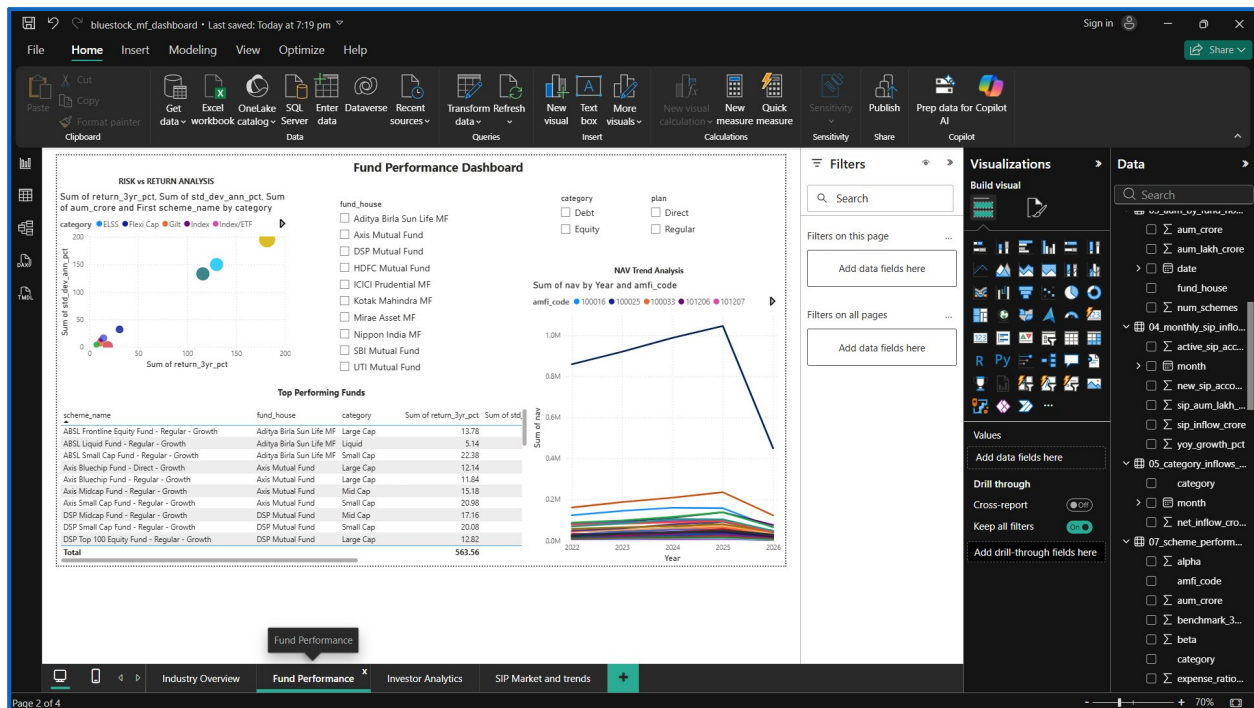


Figure: Page 2 — Fund Performance — Power BI Dashboard

Page 3 — Investor Analytics

A state-wise horizontal bar chart ranks states by total transaction amount (INR), with Punjab and Tamil Nadu leading. A donut chart shows transaction type split: Lumpsum 58.49%, SIP 35.34%, Redemption 6.17%. Average investment by age group bars reveal the 46–55 cohort investing the most per transaction. A monthly transaction volume line chart shows seasonal participation patterns. City tier and state filters enable geographic segmentation.

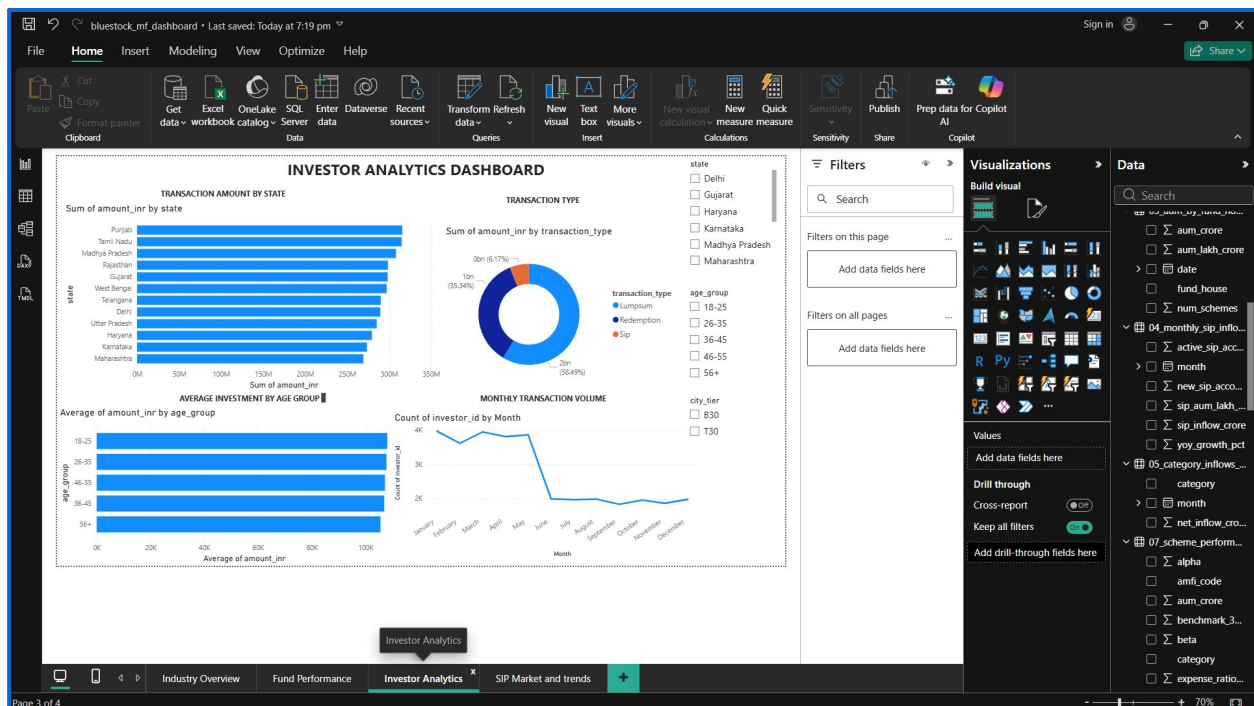


Figure: Page 3 — Investor Analytics — Power BI Dashboard

Page 4 — SIP & Market Trends

A bar chart highlights the Top 5 Categories by Net Inflow, with Liquid funds leading significantly. A Category Inflow Heatmap shows monthly inflow values across ELSS, Flexi Cap, Gilt, Hybrid, Large & Mid Cap, Large Cap, and Liquid categories. Two trend lines — SIP Inflows vs Market Close Value — demonstrate the sustained growth in both retail participation and market performance from 2022 to 2026.

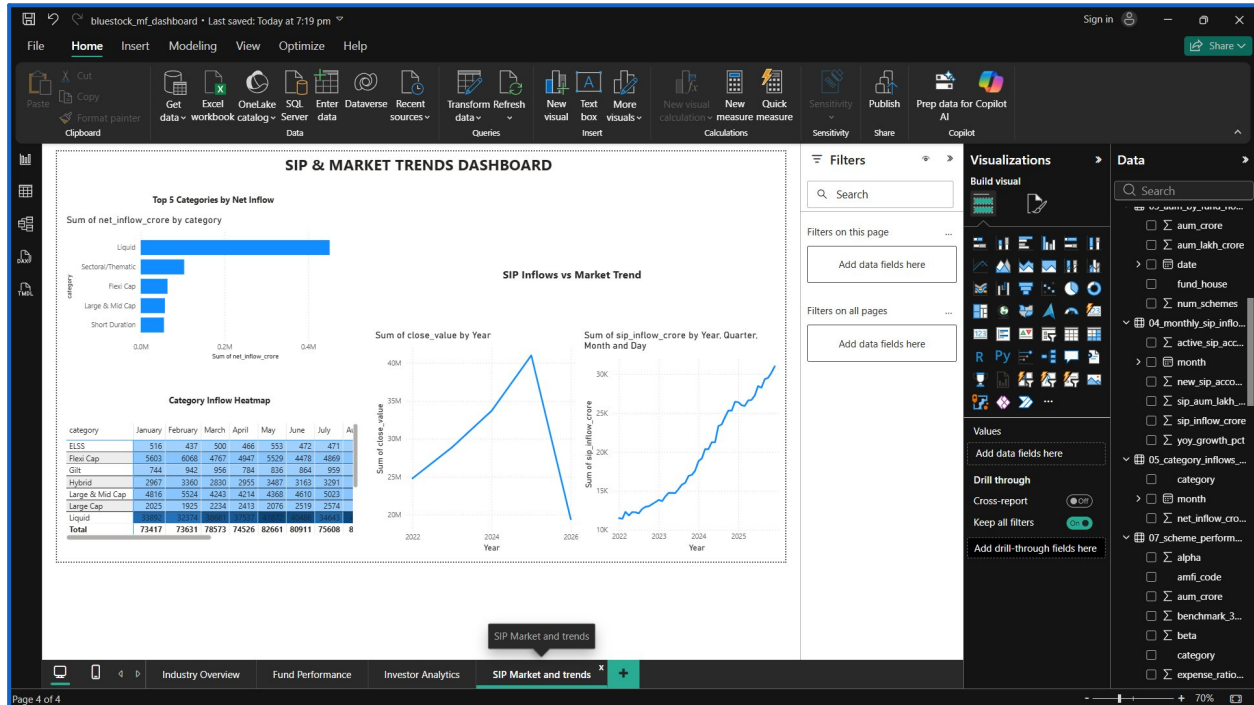


Figure: Page 4 — SIP & Market Trends — Power BI Dashboard

9. Key Findings & Recommendations

9.1 Key Findings

1	Industry Leadership: SBI Mutual Fund maintained the highest AUM throughout the study period, reflecting strong brand trust and distribution reach across retail and institutional segments.
2	NAV Growth & Correction Cycles: Equity fund NAVs demonstrated strong appreciation during the 2023 bull run, followed by a notable correction phase in 2024, consistent with broader market movements.
3	SIP Growth Trajectory: Monthly SIP inflows exhibited a sustained upward trajectory from 2022–2025, with the highest inflow period occurring in H2 2024 — reflecting deepening retail investment habits.
4	Small-Cap Risk Profile: Small-cap funds (SBI, Axis, ABSL) carried the highest downside risk as measured by VaR (95%) and CVaR (95%), consistent with their investment mandates targeting high-volatility growth securities.
5	SIP Continuity Gap: Only 2.20% of eligible SIP investors maintained regular contribution intervals below 35 days, signalling widespread SIP discontinuity and a significant retention challenge for fund houses.
6	Mid-Cap Alpha Generation: Mid-cap and emerging equity funds — including Kotak Emerging Equity and ICICI Prudential Midcap — demonstrated the strongest risk-adjusted alpha relative to the NIFTY100 benchmark.
7	Geographic Expansion: While T30 cities dominated total investment volumes, B30 city participation showed accelerating growth, indicating a broadening geographic base for mutual fund adoption.
8	Portfolio Concentration: Axis Bluechip Fund showed the highest portfolio concentration by HHI, while diversified equity and flexi-cap funds exhibited lower concentration scores.

9.2 Recommendations

→ For Fund Houses: Implement proactive SIP continuity nudge campaigns. Only 2.20% of investors maintained regular SIP intervals — targeted reminders and auto-debit improvements could significantly improve SIP retention rates.
→ For Investors — Risk-Aware Selection: Use the composite fund scorecard and VaR analysis as a multi-factor screening framework. Investors with moderate risk appetite should favour funds with Sharpe Ratios above 1.0 and maximum drawdown below 15%.

→ **For B30 Market Expansion:** The accelerating growth in B30 city participation warrants targeted distribution investment. Fund houses should prioritise digital-first onboarding and vernacular investor education campaigns in Tier-2 and Tier-3 markets.

→ **For Portfolio Construction:** The HHI and correlation analysis indicate meaningful diversification benefits from combining large-cap and small-cap funds. Investors should avoid over-concentration in high-HHI funds, particularly in volatile market conditions.

→ **For Dashboard Users:** The Power BI dashboard should be refreshed quarterly with updated NAV, AUM, and transaction data. The benchmark comparison page should be expanded to include sector-specific indices as the dataset grows.

10. Project Limitations

While the project delivers a comprehensive analytics solution, the following limitations should be considered when interpreting findings:

1. Historical Data Window

The NAV history dataset covers 2022–2026, limiting the ability to compute genuine 5-Year CAGR from historical records. ARIMA-forecasted CAGR values were generated as proxies but carry inherent model uncertainty.

2. Simulated Transaction Data

The investor transactions dataset is synthetically generated for capstone purposes and does not represent actual investor behaviour from live production systems. Real-world patterns may differ in scale, seasonality, and geographic distribution.

3. Benchmark Scope

Benchmark analysis was limited to NIFTY50 and NIFTY100. Sector-specific fund categories would be more accurately assessed against their respective sectoral indices (e.g., NIFTY Bank, NIFTY IT).

4. Risk-Free Rate Assumption

A fixed risk-free rate of 6.5% was assumed for Sharpe and Sortino ratio calculations. In practice, this rate fluctuates with RBI policy decisions and should be dynamically updated for production use.

5. ARIMA Forecasting Accuracy

Time-series forecasts are directional indicators subject to model assumptions regarding stationarity and lag selection. Actual NAV trajectories depend on macroeconomic conditions, fund manager decisions, and market events beyond the model's scope.

6. Dashboard Data Refresh

The Power BI dashboard was built on a static data snapshot. A production deployment would require scheduled data refresh integration with live data sources or an updated database pipeline.

7. SIP Continuity Threshold

The 35-day gap threshold used to flag At-Risk SIP investors is a heuristic defined for this analysis. Actual risk thresholds would need calibration against fund house operational data.

11. Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully demonstrated the complete lifecycle of a professional data analytics project — from raw data ingestion and pipeline engineering through advanced quantitative modelling and stakeholder-ready business intelligence reporting.

Over seven structured phases, the project transformed ten raw datasets and over 46,000 NAV observations into a fully validated, analytics-ready SQLite database, a composite fund performance scorecard, advanced risk metrics (VaR, CVaR, Rolling Sharpe, HHI), and an interactive four-page Power BI dashboard. A rule-based fund recommendation engine was developed to translate quantitative findings into investor-facing guidance.

The project reinforced practical competencies across the full data engineering and analytics stack — Python, SQL, SQLite, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, DAX, and ARIMA modelling — while delivering a scalable, documented, and production-ready analytics solution for the mutual fund domain.

All eight project objectives achieved. All seven deliverables completed. Pipeline executes end-to-end. Dashboard loads successfully. Repository tagged v1.0 and production-ready.

Kevin Joel Velevela BCA – Data Science velevelakevinjoel@gmail.com Bluestock Fintech Capstone 2025	Tech Stack Python · Pandas · NumPy SQLite · SQLAlchemy · Power BI Matplotlib · Seaborn · Plotly
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